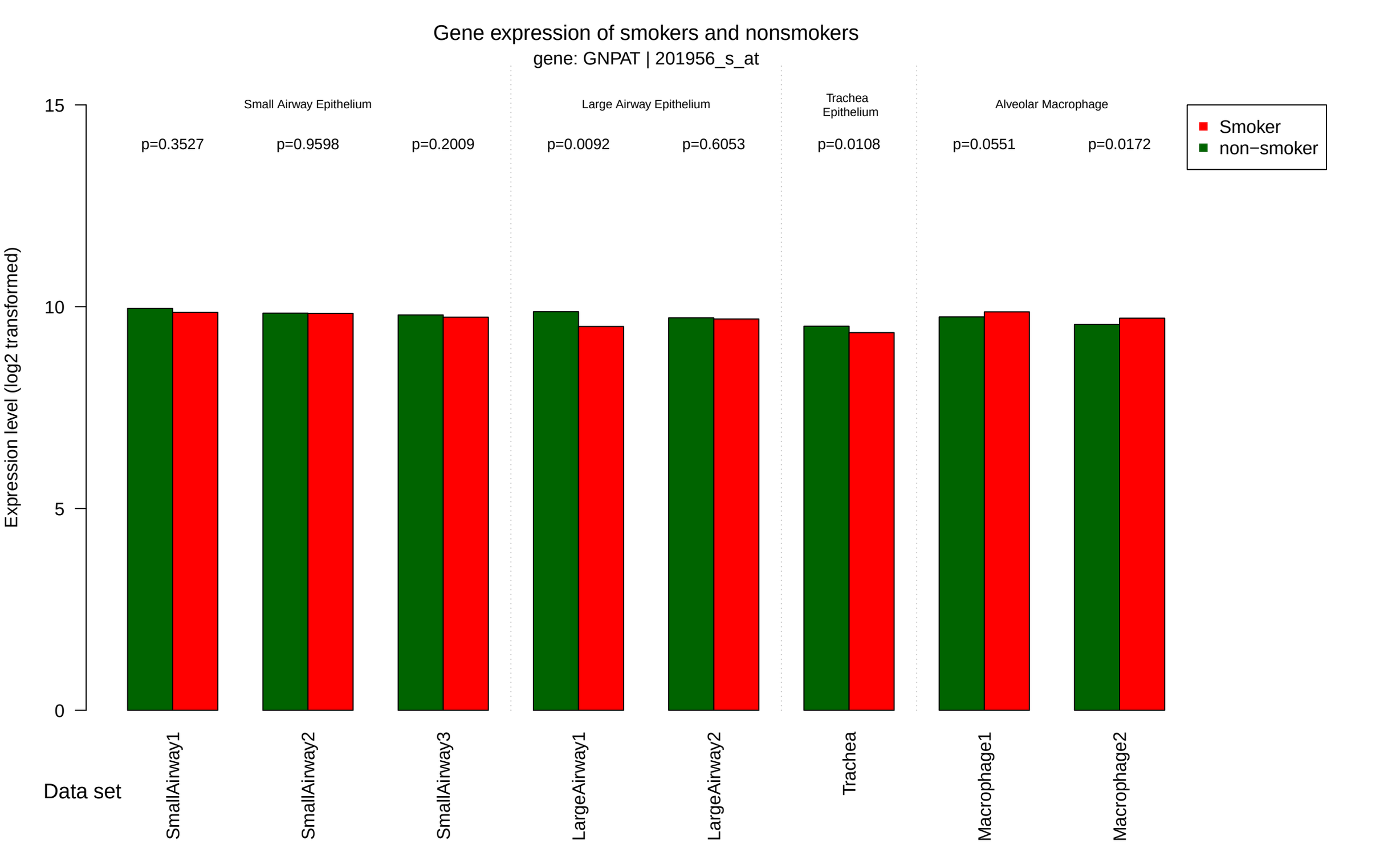
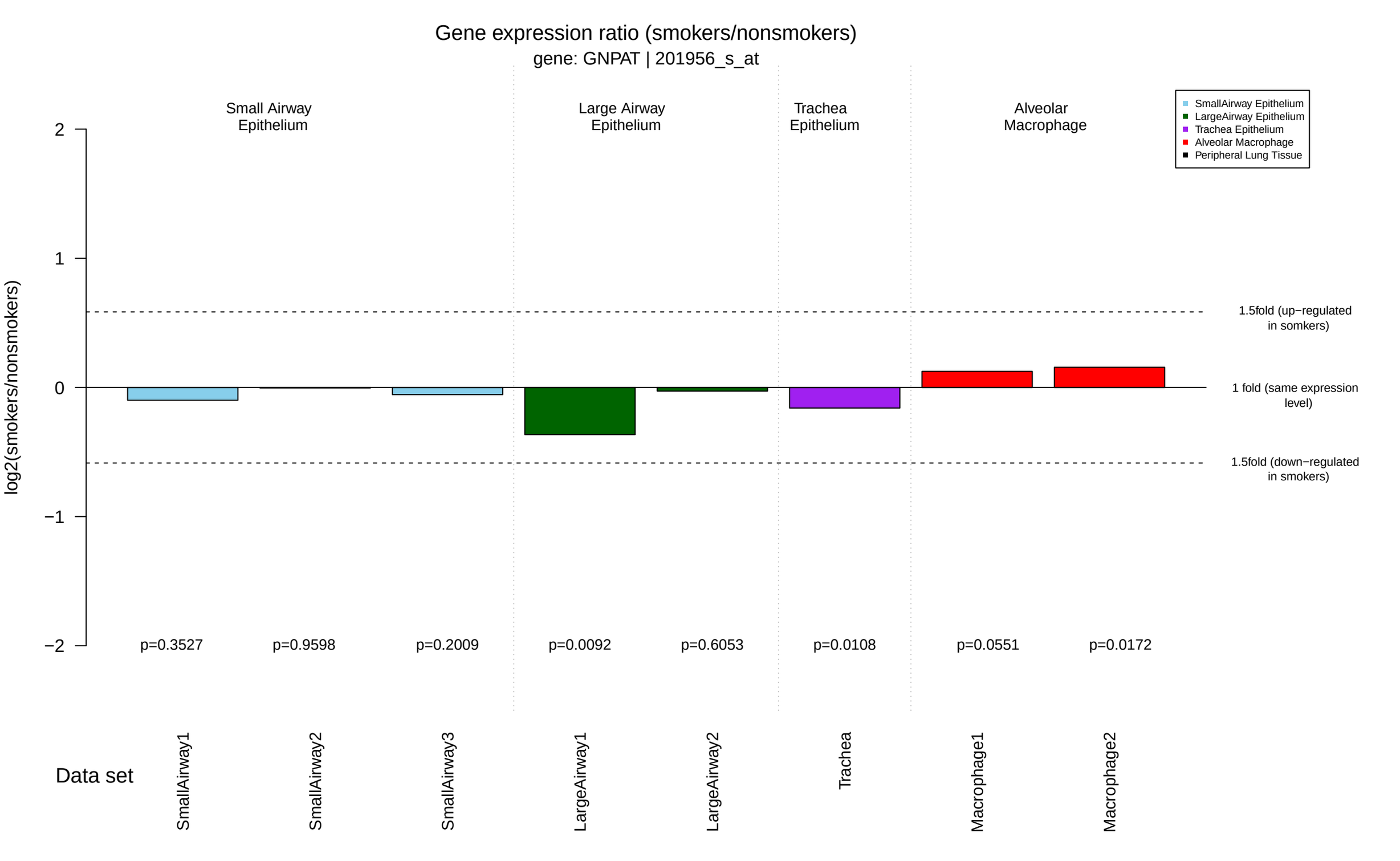
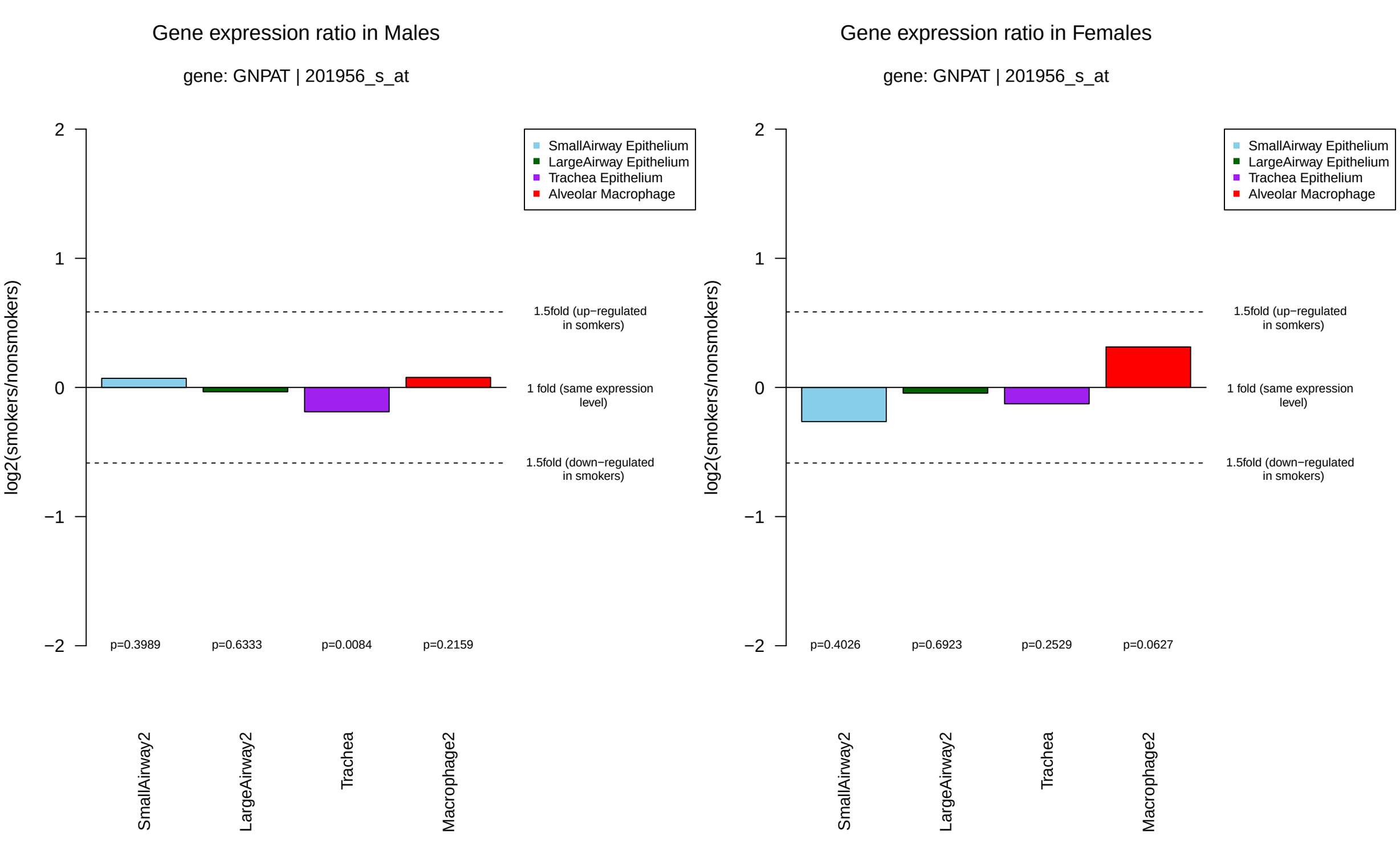
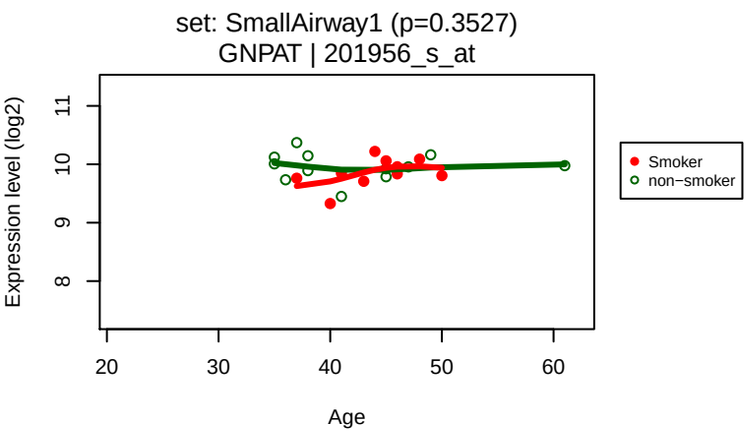
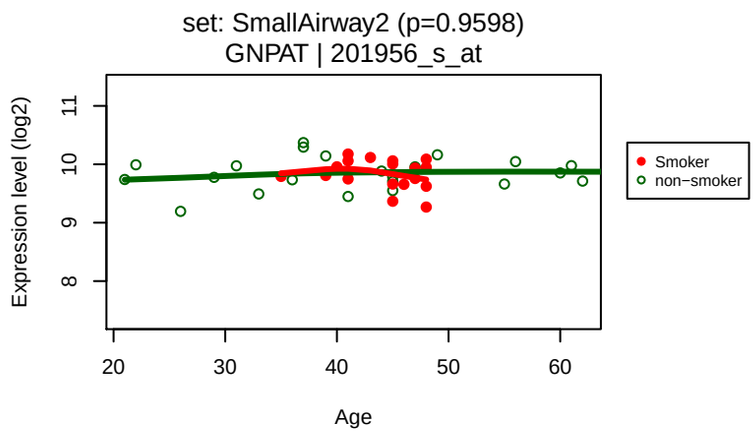
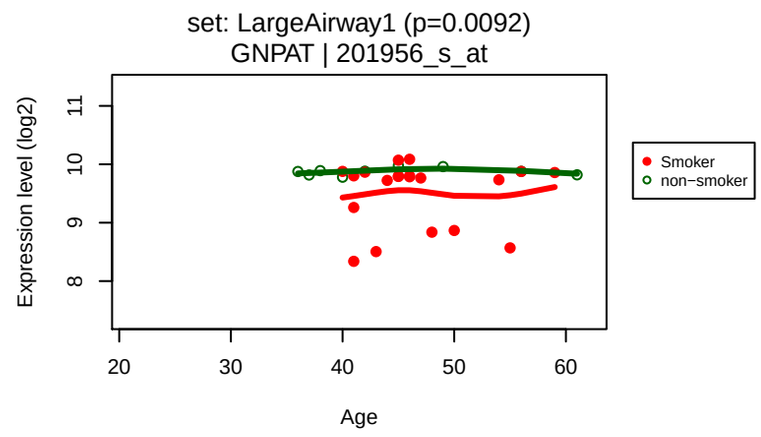

Gene expression of smokers and nonsmokers
gene: GNPAT | 201956_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3527
p=0.9598
p=0.2009
p=0.0092
p=0.6053
p=0.0108
p=0.0551
p=0.0172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

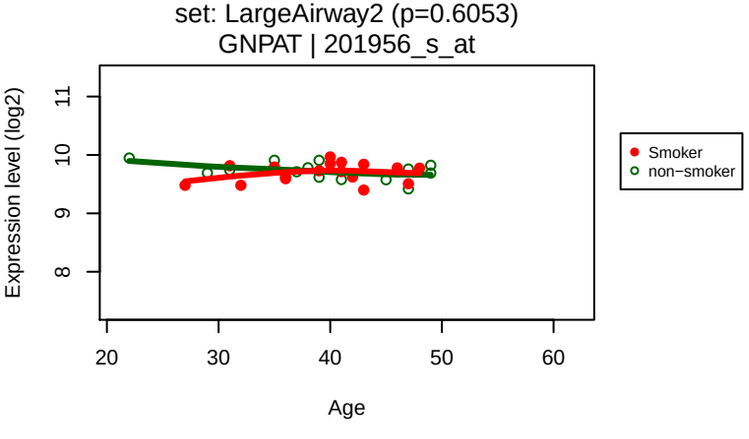

Gene expression ratio (smokers/nonsmokers)
gene: GNPAT | 201956_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3527
p=0.9598
p=0.2009
p=0.0092
p=0.6053
p=0.0108
p=0.0551
p=0.0172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

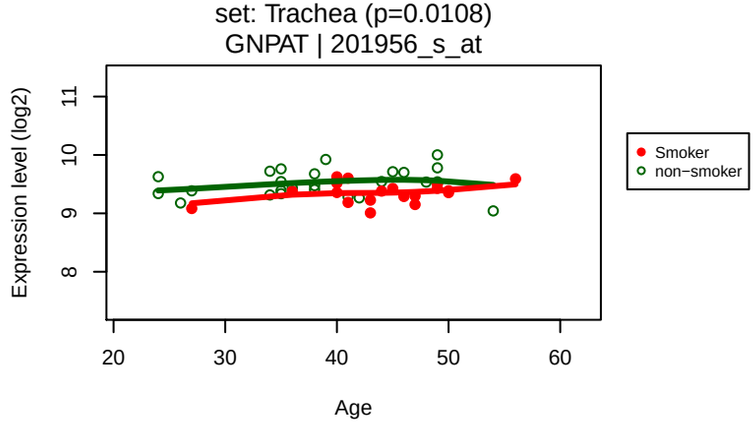

Gene expression ratio in Males
gene: GNPAT | 201956_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3989
p=0.6333
p=0.0084
p=0.2159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNPAT | 201956_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4026
p=0.6923
p=0.2529
p=0.0627
SmallAirway2
LargeAirway2
Trachea
Macrophage2

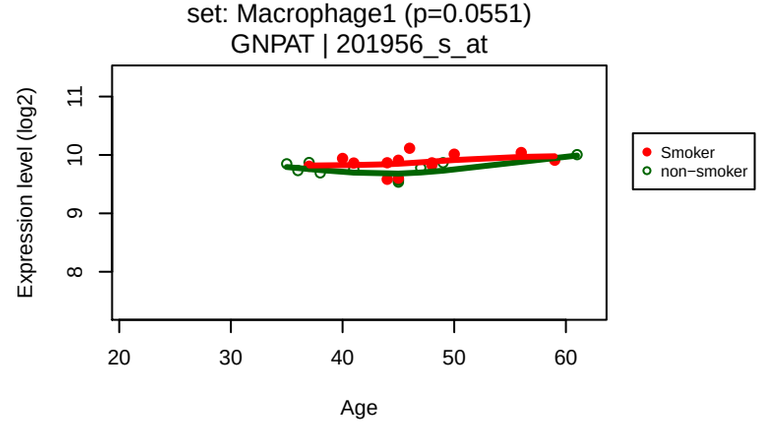

set: SmallAirway1 (p=0.3527)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker

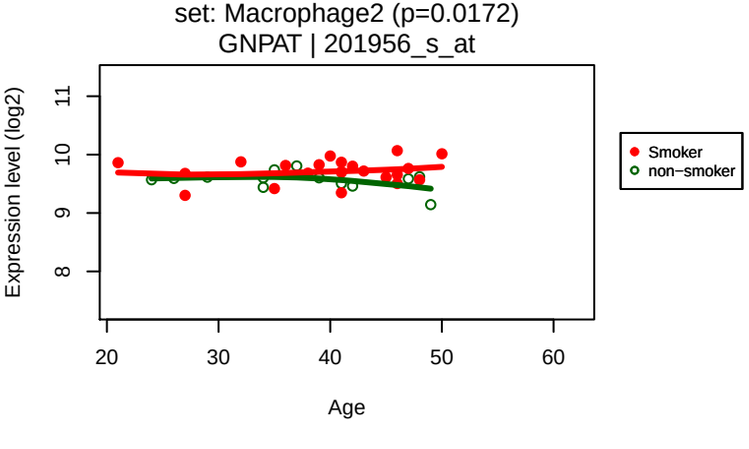

set: SmallAirway2 (p=0.9598)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker

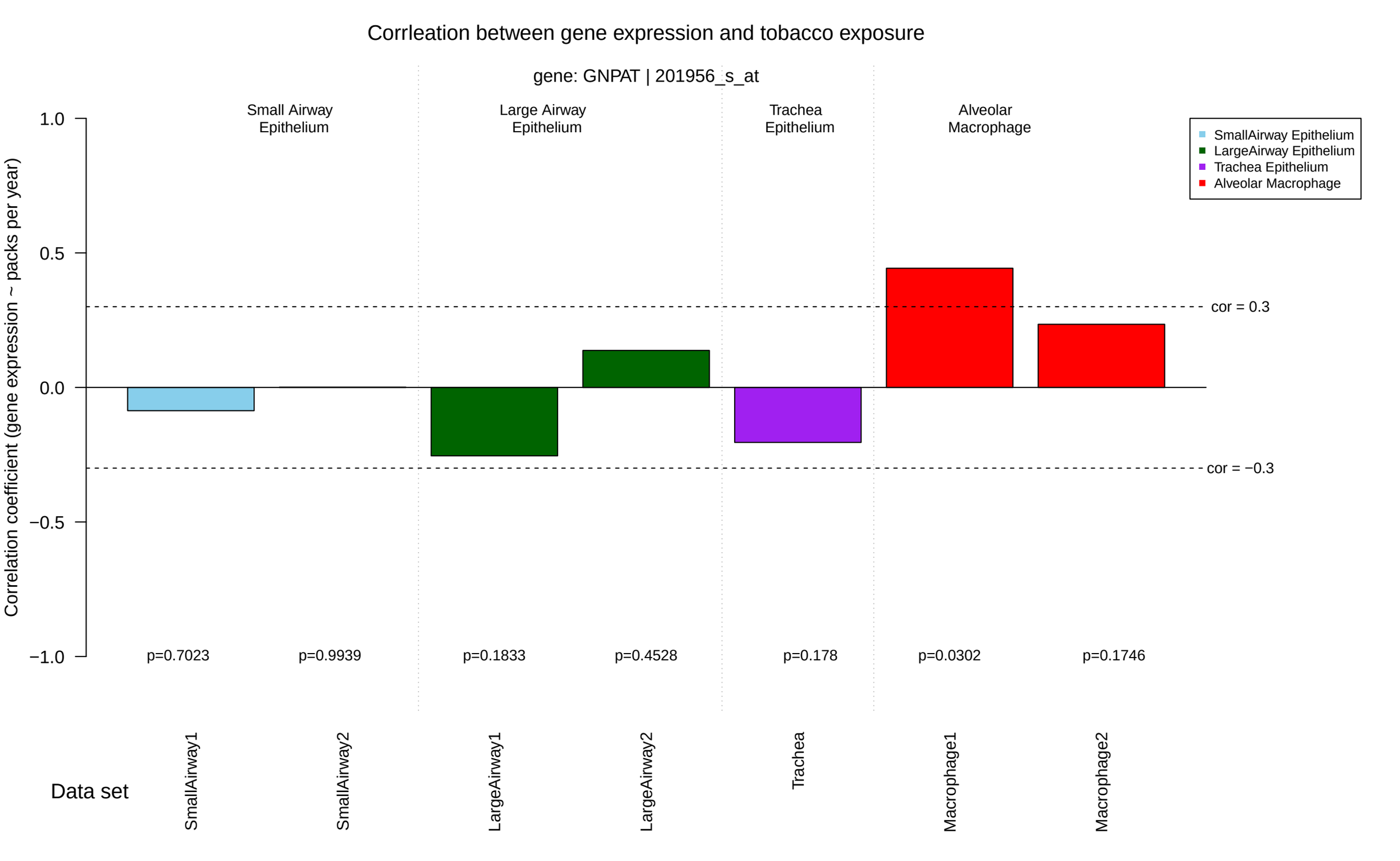

set: LargeAirway1 (p=0.0092)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6053)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0108)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0551)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0172)
GNPAT | 201956_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNPAT | 201956_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7023
p=0.9939
p=0.1833
p=0.4528
p=0.178
p=0.0302
p=0.1746
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2